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RenovAÍ: a generative artificial intelligence platform for residential renovation visualisation: a computational solution and business plan for the Brazilian market

SÃO PAULO
2026

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Final Course Project submitted to the Institute of Technology and Leadership (INTELI) as a partial requirement to obtain a Bachelor's degree in Computer Science.

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Coadvisor (if applicable): not applicable

SÃO PAULO
2026

Cataloging in Publication
Library and Documentation Service
Institute of Technology and Leadership (INTELLI)
Data entered by the author.

Sobrenome, Nome
Título do trabalho: subtítulo / Nome Sobrenome do autor; Nome e Sobrenome do orientador. – São Paulo, 2025.
nº de páginas : il.

Trabalho de Conclusão de Curso (Graduação) – Curso de [Ciência da Computação] [Engenharia de Software] [Engenharia de Hardware] [Sistema de Informação] / Instituto de Tecnologia e Liderança.
Bibliografia

1. [Assunto A]. 2. [Assunto B]. 3. [Assunto C].

CDD. 23. ed.

ACKNOWLEDGEMENTS

First and foremost, I would like to thank my advisor, Prof. Murilo Zanini de Carvalho, for his guidance, trust, and encouragement throughout this project. Even when the idea still seemed like a cloud, unclear, turbulent, and without a defined direction, he believed in my potential and helped me find alternative paths. He encouraged me not to fear the time spent studying, researching, and preparing before execution, reminding me that, many times, we must first build a strong foundation before attempting to raise a skyscraper. Beyond his role as an advisor, he became an example of passion for technology, intellectual rigour, and dedication to research. In every discussion, he showed me that arguments should prevail over emotion, and that good ideas must be built, questioned, and defended with clarity. Above all, he was an example of what it means to be a teacher, a mentor, and a true master.

I would also like to thank my older brother, Gustavo, who is my greatest example in life and one of my greatest professional role models. He battles every day for what he dreams of and spares no effort to reach the places he wants to reach. Throughout this project, he advised me, challenged my assumptions, brought new insights, and created productive conflicts of ideas that pushed me to take RenovAÍ to another level. More than that, he has always been an example of dedication, excellence, and responsibility, consistently delivering beyond what was expected and doing everything with remarkable care, precision, and commitment.

I am deeply grateful to my grandmother, Amalia. Since I was a child, she has supported me in my studies and in life, always encouraging me to learn, research, question, and pursue knowledge. She taught me to value science, curiosity, and education not only as academic tools, but as ways of understanding and transforming the world. She is also one of my greatest academic examples: at a time when many believed that women should not pursue higher education, she challenged the expectations of her generation and graduated from one of the most prestigious law schools in Brazil. Without her support, example, and encouragement, I would not have come to know Inteli, nor would I have gone as far as I once believed possible.

I would also like to thank my aunt, Silvia, for always supporting me and encouraging me to seek success while remaining generous, empathetic, and committed to helping others. She showed me that empathy is a strength, and that understanding another person’s pain is often the first step toward creating something meaningful. In many ways, this lesson made RenovAÍ possible.

I thank my father for supporting me throughout this journey and for teaching me to keep researching, studying, and seeking knowledge. He helped me understand the value of work, persistence, and responsibility. Through his work, he also awakened my interest in architecture and engineering, encouraging me to learn more, explore these fields, and see how ideas can become real structures, spaces, and solutions. In many ways, his example helped inspire the path that eventually led me to RenovAÍ.

I thank my mother, without whom I might never have returned to university. She has always cared deeply for my wellbeing and showed me the value of helping others. Through her example, I learned that when we teach, support, and serve others, we also learn and grow ourselves. In many ways, RenovAÍ is also about this exchange: helping users make better decisions while learning from their needs, doubts, and experiences.

I would also like to thank my master and life mentor, Alexandre Luke. With his wisdom, he taught me never to stop fighting, and helped me reflect on mission, values, goals, and the importance of thinking about others. His guidance has been a constant reminder that ambition must be connected to purpose.

I thank Paulo Marcos, my older brother figure, friend, and mentor. I remember being fourteen years old and listening to him tell me stories about his work at Cabify. Over time, those stories became lessons from different professional journeys, challenges, and adventures he lived through. Through them, he showed me that dedication is one of our greatest tools, that we must always be willing to learn, and that we do not need to know everything in order to move forward. What matters is understanding what we need to know, continuing to learn and evolve, and having the humility to ask for help from those who know more than we do. He taught me that the arrogance of believing we do not need help is one of our greatest flaws. For the past ten years, he has helped me think about business, life, goals, and work, always encouraging me to seek improvement, question my ideas, pursue knowledge, and become better without ever harming others in the process.

I would also like to thank my uncle, Henrique, who was a second father to me and the first mentor I ever had in life. I still remember being around ten or eleven years old and hearing him tell me that I could be very intelligent, but that I needed to learn how to build focus, because only through focus would I be able to achieve my goals. His words, affection, and concern stayed with me throughout the years. He was the first person to encourage me to seek the discipline necessary to reach my full potential. He taught me not to doubt myself, but also not to become arrogant, reminding me that true growth requires humility, constant learning, and the willingness to keep improving.

I thank the Institute of Technology and Leadership (Inteli) for the education, environment, and opportunities that made this work possible. I also thank the professors who supported me throughout this journey, from the most ambitious and unconventional ideas to the more tangible projects. Each of them, in different ways, helped me understand the value of research, discipline, and intellectual rigour.

I thank the early users of RenovAÍ, whose feedback, doubts, and reactions helped shape the product and made this project more grounded in real needs.

Finally, I thank my friends, both those who have been with me for years and those whom life brought closer to me in recent months. Some believed in this idea from its earliest and most uncertain moments, while others arrived in the final stretch and helped me not only keep going, but also aim higher, deliver more, and believe more deeply in what I was building. Their support, encouragement, ideas, and trust in me became a source of strength during one of the most demanding phases of this journey. More than simply helping me continue, they brought me happiness, and that happiness became fuel to keep fighting with even more determination, courage, and purpose.

“Recognising the need is the primary condition for design.” (Charles Eames)

ABSTRACT

WASSERSTEIN, Bruno. RenovAÍ: a generative artificial intelligence platform for residential renovation visualisation. 2026. 62 p. Final Course Project (Bachelor) – Computer Science, Institute of Technology and Leadership (INTELI), São Paulo, 2026.
RenovAÍ is a Brazilian generative artificial intelligence platform for residential renovation planning, aimed at the general public rather than professional architects. From a description or photograph of a room, the platform generates a photorealistic image of the renovated space and a list of suggested furniture, in minutes and at no cost. This work presents the development of the computational solution, based on modern web technologies and cloud generative-AI services, and the corresponding business plan. The problem addressed is uncertainty before spending: 82% of Brazilian homes are built without a qualified professional (CAU/Datafolha, 2022), and an architecture project costs from R$ 55 to R$ 140 per square metre. The business model is based on commissions on sales, with optional paid plans. Technical validation was carried out with real users in production. The solution is found to demonstrate technical feasibility and initial robustness, and the model to be promising, contingent on validating the conversion hypotheses with proprietary data.
Keywords: generative artificial intelligence; interior design; residential renovation; business plan; SaaS platform.

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LIST OF ABBREVIATIONS AND ACRONYMS

AI Artificial Intelligence

MVP Minimum Viable Product

TAM/SAM/SOM Total / Serviceable Available / Serviceable Obtainable Market

BMC Business Model Canvas

CDN Content Delivery Network

LGPD Brazilian General Data Protection Law

SWOT Strengths, Weaknesses, Opportunities and Threats

SUMMARY

1 Introduction

The home is one of the most significant spaces in a person's life: it is where daily routines unfold, where families grow and where a large share of household wealth is invested. Yet the moment of renovating, decorating or simply reorganising a room is, for most people, surprisingly intimidating. The decision mixes aesthetic taste, financial constraint, spatial uncertainty and the fear of an irreversible mistake: a sofa that does not fit, a colour that clashes, money spent on furniture that, once assembled, looks nothing like what was imagined.

Consider a common situation: a person receives the keys to a small flat, browses dozens of inspiration images on social media, and still cannot picture how any of them would look in their own living room. They do not have the budget to hire an architect for what is, in essence, a light renovation, and they are not confident enough to commit thousands of reais to furniture on the basis of imagination alone. This gap, between the abundance of inspiration and the scarcity of decision-making confidence, is the starting point of this work.

Recent advances in generative artificial intelligence have made it technically feasible, for the first time, to turn a photograph and a short description of a room into a photorealistic image of the renovated space in seconds and at negligible marginal cost. RenovAÍ is a Brazilian platform built on this possibility. This chapter introduces the context and motivation of the work, defines the problem and the value proposition, states the objectives, presents the justification and contributions, and describes the structure of the document.

1.1 Context and Motivation

Brazil has a large and active home-improvement market, sustained by structural housing conditions. The home-goods sector moved approximately R$ 102 billion in 2024 (ABCasa/IEMI, 2024). At the same time, the country has a significant housing deficit and a large stock of dwellings with some form of qualitative inadequacy (FJP, 2024), and an estimated 82% of homes are built or altered without a qualified professional (CAU/Datafolha, 2022). The result is a vast population that renovates, decorates and adapts the home it already has, largely on its own and without specialised guidance.

For this population, hiring an architect or interior designer is often economically unviable for a light renovation: a residential architecture project costs, on average, between R$ 55 and R$ 140 per square metre (GetNinjas, 2024; Habitissimo, 2024), a sum that is difficult to justify when the goal is to repaint a room, replace furniture or reorganise a space. The motivation of this work is precisely this underserved middle ground: people who want more confidence than an unaided online search can give them, but who will not (and arguably should not need to) commission a full professional project.

The technological motivation is equally important. Until recently, producing a realistic visualisation of a renovated room required either manual 3D modelling by a specialist or expensive rendering software. The maturation of generative image models has collapsed this cost, making instant, photorealistic and low-cost visualisation accessible to the general public through a mobile phone. RenovAÍ is positioned at the intersection of a large, underserved market and a newly viable technology.

1.2 Problem Definition and Value Proposition

The central problem this work addresses is the uncertainty that precedes spending on a renovation. The user can find inspiration easily, but cannot translate it into a confident decision about their own space: they do not know whether a given style, colour or piece of furniture will work, how much it will cost, or where to buy it. This translation gap between inspiration and decision generates hesitation, postponed purchases and, frequently, regret after spending.

RenovAÍ's value proposition is to deliver visual certainty before spending. From a description or a photograph of a room, the platform generates a photorealistic image of the renovated environment and a list of suggested furniture linked to real products, in minutes and at no cost to the user. The proposition combines three elements that, in the Brazilian market, are rarely found together: instantaneous AI generation, deep localisation (prices in reais, Brazilian brands, native Portuguese) and a direct path from visualisation to purchase.

It is essential to delimit the scope of this proposition. RenovAÍ acts in light renovation, decoration, furniture selection, painting, spatial organisation and preliminary visualisation. It does not perform structural calculations, does not certify the feasibility of building works, and does not replace the work of an architect, an interior designer or a civil engineer. The generated image is a tool for decision support and communication of intent, not a technical or executive project. This boundary is a deliberate product and ethical decision, and it recurs throughout the work.

1.3 Objectives

This section states the general objective of the work and the specific objectives that operationalise it.

1.3.1 General Objective

The general objective is to develop and validate a computational solution for AI-assisted residential renovation and to elaborate a business plan for its introduction into the market.

1.3.2 Specific Objectives

To achieve the general objective, four specific objectives were defined: to build a functional MVP that generates a photorealistic visualisation and a furniture list from a user briefing; to validate the product technically with real users in production; to define a revenue model for the solution; and to elaborate a complete business plan, including market sizing, competitive analysis and financial feasibility.

1.4 Justification and Contributions

The justification of the work rests on three pillars: economic relevance, social impact and technical contribution. Economically, the addressable market, annual spending by Brazilian urban households on home goods, is estimated at around R$ 92 billion (developed in Chapter 3), and the underlying behaviour, self-directed light renovation, is widespread and recurrent.

Socially, the solution has documented potential aligned with the United Nations Sustainable Development Goals (ONU, 2015), specifically Goals 1 (No Poverty), 10 (Reduced Inequalities) and 11 (Sustainable Cities and Communities), by lowering the cost of good design decisions for low-income households that improve the home they already have, rather than assuming new construction.

Technically, the contribution is a production-grade, fully localised generative pipeline for the Brazilian market, combining multi-model image generation with a furniture-recommendation layer, validated with real users in production. The work also contributes an honest, documented account of what was built, what was prototyped and what was discarded in a single-founder, AI-assisted engineering context.

1.5 Structure of the Work

This document is organised into five chapters. Chapter 1 introduces the problem, the motivation and the objectives. Chapter 2 establishes the theoretical foundation, covering generative artificial intelligence, its application to visualisation and interior design, its role in marketing and the purchase journey, the state of the art, market-sizing frameworks and digital business models. Chapter 3 develops the solution: it defines the market assumptions and hypotheses, sizes and analyses the market, examines the competition, and details the development methodology and the technological solution actually built and running in production. Chapter 4 presents the business plan, the validation carried out with real users and the results obtained. Chapter 5 concludes the work and outlines its limitations and future directions. The References, Appendices and Annexes close the document.

2 Theoretical Foundation

This chapter establishes the conceptual basis on which RenovAÍ is built and analysed. It begins with generative artificial intelligence, the core technology of the product (2.1), and its specific application to the visualisation of interior spaces, including its limitations (2.2). It then situates artificial intelligence within marketing and the consumer purchase journey (2.3), reviews the state of the art and existing tools (2.4), and finally presents the analytical frameworks used in the business plan: market-sizing models (2.5) and digital business models (2.6).

2.1 Generative Artificial Intelligence

Artificial intelligence (AI) is the field concerned with building systems capable of performing tasks that normally require human intelligence, such as perception, reasoning and decision-making. Within AI, machine learning builds models that learn patterns from data rather than following explicitly programmed rules; the recent surge in these capabilities is largely attributable to deep learning, the use of multi-layer neural networks trained on large datasets (LECUN; BENGIO; HINTON, 2015). A useful distinction separates discriminative models, which learn to map an input to a label or value (for example, classifying an image or predicting a price) from generative models, which learn the underlying distribution of the data and can produce new examples that resemble it: new text, new images or new combinations of items.

Generative AI is the branch dedicated to this second capability. Three families of architectures underpin the current generation of generative systems. Generative adversarial networks pose generation as a game between a generator and a discriminator that compete until the generated samples become indistinguishable from real ones (GOODFELLOW et al., 2014). The Transformer architecture, based on the attention mechanism, enabled large language models capable of generating coherent text and interpreting natural-language instructions (VASWANI et al., 2017). Diffusion models, which learn to reverse a gradual noising process, became the state of the art for high-fidelity image synthesis (HO; JAIN; ABBEEL, 2020), and latent diffusion models made this synthesis computationally efficient enough for widespread use (ROMBACH et al., 2022).

For the purposes of this work, three generative modalities are relevant. Text generation, via large language models such as the GPT family (BROWN et al., 2020) and bidirectional encoders such as BERT (DEVLIN et al., 2019), allows the system to interpret a user's briefing, ask structured questions and produce descriptions and prompts. Image generation, via diffusion models and multimodal systems such as CLIP (RADFORD et al., 2021), DALL-E 2 (RAMESH et al., 2022) and Imagen (SAHARIA et al., 2022), produces the photorealistic visualisation of the renovated room. Recommendation, which may combine generative and discriminative techniques with semantic embeddings learned from data (MIKOLOV et al., 2013), connects the generated scene to real furniture and products. RenovAÍ orchestrates these three modalities into a single user-facing flow.

Table 1 – Traditional (discriminative) AI versus generative AI

Aspect	Traditional / discriminative AI	Generative AI
Primary task	Classify, predict, rank	Produce new content
Typical output	A label, a score, a category	Text, image, audio, combinations
Example in this domain	Classifying a room style	Generating an image of the renovated room
Main risk	Misclassification	Hallucination and physically implausible output

Source: prepared by the author (2026).

2.2 Generative AI in Visualisation and Interior Design

Applied to interior spaces, generative image models reduce a specific and costly form of uncertainty: the difficulty most people have in mentally projecting how a change will look before it is made. By producing a concrete, photorealistic image from a photograph and a short description, the technology turns a diffuse intention into a sharable visual reference, supporting decisions about style, colour, furniture and spatial arrangement. This uncertainty-reduction function is the core mechanism through which RenovAÍ creates value.

However, generative visualisation has intrinsic limitations that must be stated explicitly and that shape the product's design and its honest positioning. Image models can hallucinate, inventing objects, textures or architectural elements that were not requested and do not exist. They frequently make errors of scale and proportion, rendering furniture at sizes that would not fit the real room. They can produce geometric inconsistencies, such as impossible perspectives, distorted straight lines or incoherent reflections. And they struggle with coherence across iterations: asking for a small change can unexpectedly alter the whole scene, making controlled, incremental editing difficult.

These limitations carry a question of technical responsibility. A generated image is a plausible visual suggestion, not a measurement, a technical drawing or a guarantee of feasibility. It cannot certify that a wall may be removed, that a structure will bear a load or that a measurement is exact. For this reason, RenovAÍ is deliberately scoped to light renovation, decoration, furniture selection, painting, spatial organisation and preliminary visualisation, and it does not replace the work of an architect, an interior designer or a civil engineer. Any intervention with structural, electrical or hydraulic implications requires a qualified professional. This boundary is treated, in this work, as both an engineering constraint and an ethical commitment.

Table 2 – Generative visualisation: capabilities and limitations

Capability	Corresponding limitation	Product mitigation
Instant photorealistic image	Possible hallucination of elements	Guided briefing; framing as a suggestion, not a project
Style and colour exploration	Errors of scale and proportion	Preservation of the original room; scope limited to light renovation
Low marginal cost	Geometric inconsistencies	Version history so users can compare and discard
Accessible via mobile phone	Limited coherence across edits	Iterative refinement treated as an open research problem

Source: prepared by the author (2026).

2.3 Artificial Intelligence in Marketing and the Purchase Journey

The consumer purchase journey is commonly described as a progression from awareness to consideration to decision and, finally, to purchase and post-purchase (KOTLER; KELLER, 2012). In categories with high aesthetic and financial involvement, such as furniture and home renovation, the consideration stage is long and uncertain: the consumer accumulates inspiration but hesitates to commit, because they cannot easily evaluate how a product will look and perform in their own context. Friction at this stage is a primary cause of abandoned or postponed purchases.

Artificial intelligence intervenes in this journey in two complementary ways. First, through personalisation and recommendation, matching products to a user's expressed preferences and context. Second, and more distinctively for this work, through generative visualisation, which allows the consumer to see the product applied to their own environment before buying. This visual-commerce mechanism shortens the path from inspiration to decision by replacing imagination with a concrete image, reducing perceived risk at the exact moment of greatest hesitation.

RenovAÍ is designed around this insight: the generated image is not only an aesthetic deliverable but a conversion instrument, positioned at the consideration stage to convert diffuse inspiration into a confident purchase decision, with the suggested-furniture list providing the direct bridge to the transaction.

2.4 State of the Art and Existing Tools

The application of generative AI to interior design has moved rapidly from research to product. Internationally, several reference players define the state of the art and frame the category. Retailers have integrated generative visualisation directly into the shopping experience, as with Wayfair's Decorify and Muse tools, which generate styled rooms linked to a product catalogue (WAYFAIR, 2023). Specialised platforms such as Houzz combine a professional community, a marketplace and visualisation tools; indie products such as RoomGPT demonstrated strong demand for pure-AI room generation; and hybrid services such as Havenly combine artificial intelligence with human designers and a marketplace. The closure of Modsy, a pure-human 3D rendering service, illustrates the difficulty of scaling models that depend on manual work per project.

In the Brazilian market, a mapping conducted for this work identified both AI-only tools and human-assisted services, but none combining instant generative visualisation, deep localisation and a shoppable furniture list adapted to the country. Theoretically, competitive positioning in such a category can be read through established frameworks of industry structure (PORTER, 1980) and of how simpler, more accessible solutions tend to displace incumbents over time (CHRISTENSEN, 1997). The detailed competitive analysis is presented in Chapter 4; the product references cited above are used as market sources that document the technological state of the art, not as peer-reviewed academic literature.

2.5 Market Sizing Frameworks (TAM, SAM, SOM)

Estimating the size of a market is a standard step in evaluating a business opportunity. A widely used framework decomposes the opportunity into three nested layers. The Total Addressable Market (TAM) is the total demand for a category if a solution captured the entire relevant market. The Serviceable Available Market (SAM) restricts the TAM to the portion that the solution can realistically serve, given its segment, geography and business model. The Serviceable Obtainable Market (SOM) is the share of the SAM that the solution can plausibly obtain in a defined period, given competition and execution capacity.

Two complementary approaches are used to quantify these layers. The top-down approach starts from aggregate market figures published by industry sources and applies successive filters. The bottom-up approach starts from unit economics, the number of potential users multiplied by an average revenue per user, and aggregates upwards; it is generally considered more defensible because each factor is traceable to a primary source. In this work, the market is sized bottom-up, with each factor referenced to a published source, and the quantitative application of this framework is developed in Chapter 4.

2.6 Digital Business Models

A business model describes how an organisation creates, delivers and captures value. The Business Model Canvas is a widely adopted tool that represents a business through nine building blocks, including value proposition, customer segments, channels, revenue streams and cost structure (OSTERWALDER; PIGNEUR, 2010). It is used in this work to organise the description of the RenovAÍ business model in Chapter 4.

Several revenue mechanisms are common in digital products and are relevant to this work. In the commission or affiliate model, the platform earns a percentage on sales it originates, aligning its revenue with the value it generates for the user and keeping the core experience free. In the marketplace model, the platform connects supply and demand (for example, artisans or retailers with buyers) and monetises the transaction or the listing. In the freemium and paid-plan model, a free tier drives adoption while heavier or professional users pay for additional capacity or features. Software-as-a-Service (SaaS) delivers software over the cloud on a recurring basis, typically with low marginal cost per additional user. In early-stage ventures, the choice and tuning of these mechanisms is typically validated incrementally, in line with lean-startup and customer-development practice (RIES, 2011; BLANK; DORF, 2012).

RenovAÍ combines these mechanisms: a free, commission-funded core experience, optional paid plans for heavier users, and a future marketplace connecting local suppliers to buyers who already have a defined visual project. An overview of the business model is given in Chapter 4 and a consolidated, generic business plan in Appendix A.

3 Methodology

This chapter describes the methodology adopted to develop RenovAÍ: the development process and methods (3.1), the functional and non-functional requirements and the use cases (3.2), and the system architecture and data model (3.3). It documents how the solution was built. The results obtained, the market analysis and the validation with real users are reported in Chapter 4.

3.1 Development Process and Methods

This section documents how RenovAÍ was built: the development process and methods and the journey across roughly nine months, together with the principles that guided the technical and product decisions. The solution is a fully digital, mobile-first platform in production at renovai.ia.br.

A set of design principles guided the technical and product decisions throughout the project. The platform should guide the lay user, reducing the need for free prompting, and keep the core flow simple even when the backend is complex; it should preserve the original room and avoid any structural promise, and store versions so the user can compare alternatives. On the technical side, it should abstract the AI providers and allow fallback between them, treat the catalogue as a data system rather than mere text, and protect personal data while clearly communicating the product's limits.

Because the project was led mainly by a single founder-developer in a context of high technical uncertainty, the formal methodology was adapted to a reality of a lean team, rapid prototyping and constant technical validation. In practice the process combined elements of Scrum, informal Kanban and exploratory, prototype-driven development, with intensive use of AI assistants as engineering tools.

These methods played out concretely as follows. Academic Scrum provided organisation by sprints, with incremental deliveries and documentation by phase, while an informal Kanban kept a continuous list of tasks, bugs, improvements and hypotheses. Rapid prototyping was used to build screens, flows and agents that tested feasibility, and AI-assisted development (with intensive use of ChatGPT, Claude Code, Cursor and generative models) accelerated implementation. Validation was exploratory, through manual testing, real usage and the collection of user perception, and the architecture evolved gradually, changing backend, database, frontend and modules as the product matured.

The development did not follow an industrial process with perfect documentation from the start; it was marked by cycles of discovery, attempt, discard, rewrite and consolidation, which is coherent with a technological-innovation project, especially in generative AI, where technical uncertainty is high. The work can be organised into five phases.

Table 3 – Phases of the project

Phase	Focus	Result
1	Problem exploration	Definition of the pain, the audience and the initial proposition
2	Technical validation	Tests with language models, image generation, catalogue and agents
3	Functional MVP	Project flow, visual generation, history and website
4	Module expansion	AI Designer, personas, retailer portal, condominiums, 3D
5	Academic consolidation	Documentation, auditing and technical/narrative refinement

Source: prepared by the author (2026).

In the first phase, problem exploration, the focus was understanding the pain of an ordinary user when renovating or decorating. A central thesis emerged: the greatest pain is not a lack of inspiration, but the lack of translation between inspiration and decision. Key learnings were that the user wants confidence rather than just a pretty image; that a light renovation involves emotional, aesthetic, financial and practical decisions; that the AI must be guided by structured questions rather than free prompts; and that the solution must not look like a technical tool for professionals.

The second phase, technology validation, investigated whether textual and visual data about a room could be turned into useful proposals. It explored language models for briefing interpretation, image-generation models, specialised agents, product recommendation, catalogue search, layout generation, simplified 3D rendering, structured JSON outputs, a relational database and embeddings for semantic similarity. An initial experimental architecture (FastAPI, a standalone PostgreSQL database and Docker, with agent services) was built to validate endpoints and recommendation; a known technical issue was aligning embedding dimensionality and the consistency between the models used for indexing and search. The main learning was that isolated image generation does not solve the problem: the real challenge is connecting image, briefing, budget, style and real products, while preserving versions so the user can compare alternatives.

The third phase produced a functional MVP, turning the experiments into a usable web platform with a main flow of project creation, result visualisation and history. It was in this phase that the architecture migrated from the initial experimental backend to the current serverless stack. Important decisions were to focus on light renovation (reducing technical and legal risk); to be mobile-first (the user photographs the room with a phone); to use native Brazilian Portuguese; to store versions (renovations involve comparing alternatives); to avoid any structural promise; and to separate inspiration from execution. Learnings were that the first generated image has a strong emotional impact but must be explainable, that the history is essential because a project is not decided in a single session, and that a good interface reduces the user's dependence on prompt engineering.

Table 4 – Key decisions of the MVP phase

Decision	Justification
Focus on light renovation	Reduces technical and legal risk
Mobile-first	The user photographs the room with a phone
Native Brazilian Portuguese	A Brazilian product with accessible language
Store versions	Renovations involve comparing alternatives
Avoid any structural promise	Safety and technical responsibility
Separate inspiration from execution	The product should help decide, not only generate a pretty image

Source: prepared by the author (2026).

The fourth phase explored module and vertical expansion: after the MVP, the project investigated more ambitious modules: a conversational AI Designer, architect personas, a debate between styles, a retailer portal, a condominium vertical, a Layout Studio 3D, AR/VR experiments, a moodboard and analytics. This phase was exploratory: some ideas became functional interfaces or routes, others remained concepts. The main learning was that a multi-agent approach is useful for ideation but can add excessive complexity for a lay user, and that the product must avoid becoming a scattered collection of demos.

The fifth phase, academic consolidation, turns the project into academic and technical material: documentation of the architecture, a product audit, organisation of the modules, an analysis of testing, definition of future work and preparation of this report. This report follows the ABNT NBR 14724 (ABNT, 2011) presentation standard and NBR 6023 (ABNT, 2018) for references, using the Inteli final-project template (INTELI, 2025). The challenge is to document RenovAÍ honestly, without overstating what is still a prototype and without revealing sensitive business detail.

Beyond the code stack, detailed in Section 3.3, the project relied on a small set of tools. The main tools were ChatGPT, for ideation, documentation, prompts and technical and product decisions; Claude Code, for implementation, refactoring and code reading and auditing; and Cursor, for AI-assisted development. Version control and code organisation used Git and GitHub, while the Vercel and Supabase dashboards supported deployment, build monitoring and the management of the database, authentication and access policies; spreadsheets and CSV files were used to structure the catalogue and supporting research.

Artificial intelligence was used not only as a product feature but also as a development tool. It assisted in creating React components, analysing build errors, writing endpoints, reviewing architecture, creating seed scripts, debugging database problems, generating internal prompts, organising types and entities, writing documentation, comparing technical alternatives and planning tests.

Concretely, AI helped structure the initial backend, suggesting an organisation with experimental services, a database and agents; it helped create and standardise the CSV fields of the product catalogue; it diagnosed deployment problems involving scripts, runtime versions and build configuration; it refined image prompts to produce more coherent rooms; it structured the narrative and the technical documentation; and it supported the product audit, analysing flow, UX, inconsistencies and maturity.

AI was also used to simulate the perspectives of different users and specialists (a lay user, an architect, an interior designer, a UX auditor, an AI engineer, a retailer and a sceptical user), which allowed RenovAÍ to be evaluated from multiple angles without a full multidisciplinary team at every step.

The process also had clear limitations. The lean team reduced the capacity for continuous testing, documentation and QA, and the heavy dependence on generative AI introduced output variability that is hard to control. Spreading the product across many fronts created a risk of dispersion between modules, and the limited test automation increased the risk of regressions. Because the architecture changed over time, some older decisions are harder to trace; the dependence on external APIs carries risks of cost, availability and model changes; and the still-limited broad validation means the business hypotheses require further testing with users.

3.2 Requirements and Use Cases

3.2.1 Requirements and specifications

The system requirements were derived from the product vision and refined throughout development. The tables below list the functional and non-functional requirements with their current status (concluded, partial, prototype, planned or not implemented).

Table 5 – Functional requirements

ID	Requirement	Status
RF01	User sign-up and login	Concluded/partial
RF02	Create a renovation or decoration project	Concluded/partial
RF03	Select the room type	Concluded/partial
RF04	Collect style preferences (style, colours, atmosphere)	Concluded/partial
RF05	Upload a photo of the room	Partial/concluded
RF06	Describe the room textually, without a photo	Partial/concluded
RF07	Generate a visual proposal with AI	Concluded/partial
RF08	Save the generated result and project data	Concluded/partial
RF09	Access a history of previous projects	Concluded/partial
RF10	Organise projects by property	Partial
RF11	Organise projects by room	Partial
RF12	Create and view alternative versions of a project	Partial
RF13	Recommend furniture compatible with the room and style	Partial/prototype
RF14	Display the recommended product list with basic information	Partial
RF15	Link to external products when available	Partial/to confirm
RF16	Export a project summary/budget as a PDF	Partial/to confirm
RF17	Chat with an AI assistant	Prototype/partial
RF18	Simulate an evaluation by multiple personas	Prototype
RF19	Retailer portal for product registration/management	Planned/draft
RF20	3D spatial visualisation of the room	Prototype
RF21	Generate or present a style moodboard	Partial/planned
RF22	Record key usage events (analytics)	Partial/to confirm
RF23	Inform that the product does not replace a professional project	Partial/concluded
RF24	Handle generation errors and allow a retry	Partial/to confirm
RF25	Allow the user to delete projects	To confirm

Source: prepared by the author (2026).

Table 6 – Non-functional requirements

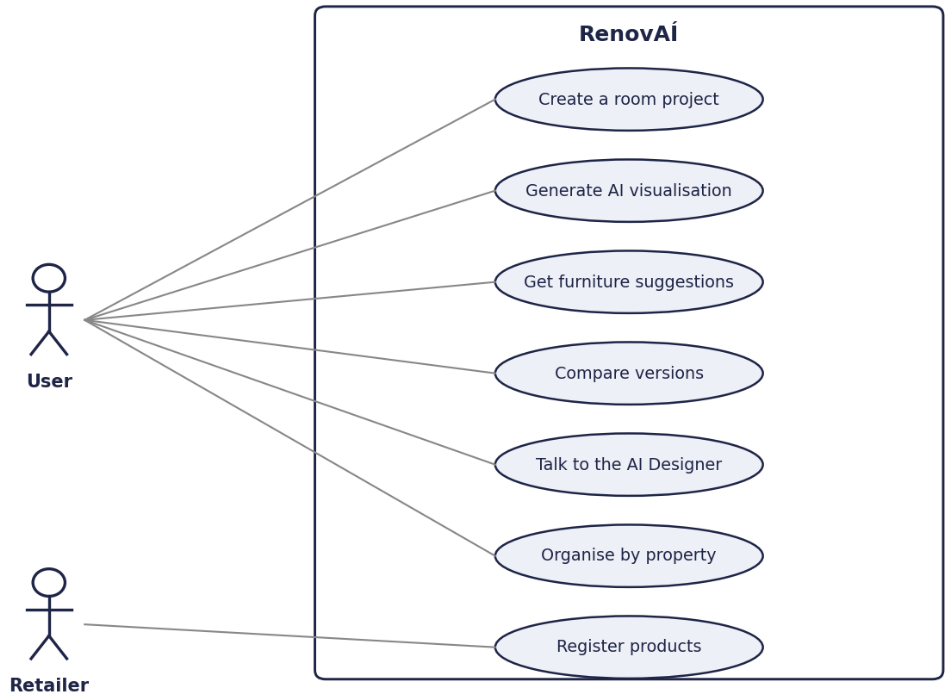
ID	Requirement	Status
RNF01	Usability for lay users, without architectural jargon	Partial/concluded
RNF02	Responsiveness on phones and desktops	Partial/concluded
RNF03	Clear loading states during long operations	Partial
RNF04	Availability on reliable hosting for public access	Partial/concluded
RNF05	Security: users cannot access other users' data	Partial/concluded
RNF06	Privacy: careful handling of home images and data	Partial/to confirm
RNF07	Scalability without a full rewrite	Partial
RNF08	Observability of failures and errors	Partial/to confirm
RNF09	Maintainability through modular, reusable code	Partial
RNF10	Reliability of AI outputs (consistent and aligned with the briefing)	Partial
RNF11	Accessibility	Planned/not formally validated
RNF12	Cross-browser compatibility	Planned/partial; formal tests not confirmed

Source: prepared by the author (2026).

3.2.2 Use cases

The main use cases describe how different actors interact with the platform. Figure 1 presents the use-case diagram, summarising how the two main actors, the end user and the retailer, interact with RenovAÍ.

Figure 1 – Use-case diagram of RenovAÍ



Source: prepared by the author (2026).

The main use cases are described below, each with its actor, the flow it follows and its current status. The first use case is the creation of a room project by an ordinary user: the user accesses the platform, logs in or starts a project, chooses the room type, enters their preferences and sends a photo or a description; the system generates the image, which the user saves and can revisit later in the history. This flow is concluded or partial.

The second use case is comparing versions of a living room, performed by an undecided user who opens an existing project, requests a new version in another style and compares the previous and the new image before deciding on a direction; this flow is partial. The third use case is receiving furniture suggestions, for a user who wants to buy items: after the room is generated, the system identifies the needed categories and recommends items, which the user reviews before following the external links; this flow is partial or at prototype stage.

The fourth use case is organising projects by property, for a user with more than one room or property, who creates a property, adds rooms, generates versions per room and consults the history organised by property; this flow is partial. The fifth use case is talking to the AI Designer when the user is unsure of what they want: the user describes the problem, the agent asks questions, the system summarises the preferences and the user generates a proposal; this flow is at prototype or partial stage.

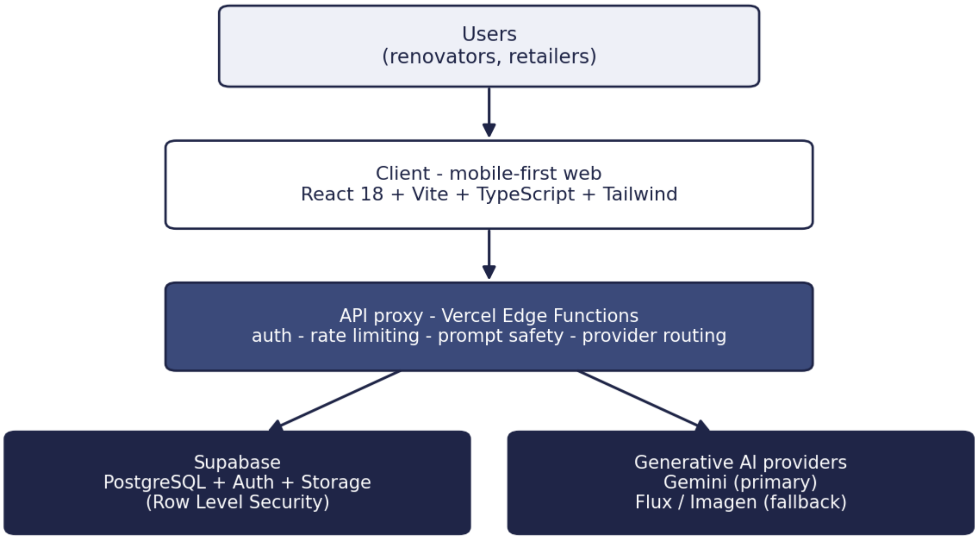
The sixth use case is evaluation by personas, for a user who wants a critical opinion: the user submits a project, the personas evaluate it and the system summarises conflicts and consensus into a final recommendation; this flow is at prototype stage. The seventh use case is a retailer registering products, in which the retailer accesses the portal, registers the store and submits a catalogue, after which the products become available for recommendation; this flow is planned or in draft. Finally, the eighth use case is a building manager viewing a common-area improvement: the manager informs the common area, the platform generates a visual proposal used for internal discussion, and the next steps are decided outside the platform; this flow is at concept or partial stage.

3.3 System Architecture and Data Model

3.3.1 System architecture

RenovAÍ is a mobile-first web application supported by services for authentication, database, storage, internal APIs and external generative-AI services. Figure 2 shows the logical architecture in layers: a web and mobile frontend; an authentication and session layer; an application layer with internal APIs; a relational database; the external AI services for image generation, briefing interpretation and recommendation; the storage of results; and the review, history and export interface.

Figure 2 – Logical architecture of RenovAÍ in layers



Source: prepared by the author (2026).

The architecture evolved over the project. An initial experimental backend (FastAPI, a standalone PostgreSQL database and Docker, with agent services) was used in the technology-validation phase. The production architecture then migrated to a serverless stack, which lowered operational complexity and cost and improved deployment speed. The current stack is summarised below.

Table 7 – Current technology stack

Layer	Technology	Status
Frontend	React 18 + TypeScript (strict) + Vite	Concluded/in production
Styling / UI	Tailwind CSS + shadcn/ui + Radix UI	Concluded/in production
State / forms	Zustand; React Hook Form + Zod	Concluded/partial
3D	Three.js + React Three Fiber	Prototype
Backend	Vercel Edge / serverless functions	Concluded/in production
Database / Auth / Storage	Supabase (PostgreSQL, Auth, Storage)	Concluded/in production
Rate limiting	Vercel KV (Upstash Redis)	Concluded/partial
Deployment	Vercel	Concluded/in production
Image generation	Google Gemini (premium model), with Imagen 4 and Flux as fallback	Concluded/partial
Text / vision AI	Google Gemini (briefing, vision, agents)	Concluded/partial
Earlier experimental backend	FastAPI + PostgreSQL + Docker	Superseded (v1)

Source: prepared by the author (2026).

The main technology choices were guided by the need for a lean, low-cost and fast-to-iterate stack suitable for a solo founder.

Table 8 – Technology choices and rationale

Choice	Rationale
React + Vite + TypeScript	A fast, strongly typed frontend with quick builds and a large ecosystem
Serverless (Vercel Edge)	No servers to manage, automatic scaling and a low fixed cost
Supabase	Database, authentication and storage with row-level security in a single managed service
Managed Redis	Rate limiting without dedicated infrastructure
Multi-model AI with fallback	Availability and resilience against a single provider's failure or price change
Mobile-first	The user photographs the room with a phone
Native Brazilian Portuguese	A product localised for the Brazilian market

Source: prepared by the author (2026).

The application is organised into layers. The presentation layer concentrates the landing page, the project-creation flow, the result screen, the logged-in area, the project history, the property/room/version pages, the AI Designer and the experimental panels. Its responsibilities are to guide the lay user, reduce the need for free prompting, display visual results, organise project data and communicate the product's limits. The application layer coordinates the business rules and AI calls: creating a project, updating the briefing, saving the original image, calling the language and image models, storing the result, creating a version, fetching the history, generating a budget and recommending furniture. The data layer stores users, properties, rooms, projects, versions, images, recommendations, products, chat messages, analytics events and generation logs.

The AI layer should be understood not as a single prompt but as a set of specialised steps. In a simplified logical pipeline, the user input is first normalised; the intent is interpreted and the constraints extracted; a structured briefing is composed and an internal prompt generated; the image model is then called, with fallback to an alternative provider; the output is post-processed and stored; the relevant items are recommended; and, finally, the result is explained to the user. The specific prompt-engineering techniques are a competitive differentiator and are intentionally kept generic in this public document.

3.3.2 Data model

The data model reflects the real structure of a residential renovation. The main entities and their status are listed below.

Table 9 – Main entities

Entity	Description	Status
User	An authenticated user	Concluded/partial
Property	A registered home	Partial/concluded
Room	A room within a property	Partial/concluded
Project	A renovation or decoration project for a room	Concluded/partial
Version	A variant of a proposal	Partial/concluded
GeneratedImage	An AI-generated image	Concluded/partial
OriginalImage	A photo uploaded by the user	Partial
Briefing	User answers and preferences	Concluded/partial
Product	A catalogue item	Partial
Recommendation	A product recommended for a project	Partial/prototype
ChatSession / ChatMessage	A conversation with the AI	Partial
AgentPersona	An architect/designer persona	Prototype
Retailer	A partner store	Planned/draft
AnalyticsEvent	A usage event	Partial/to confirm

Source: prepared by the author (2026).

The relationships follow a clear hierarchy: a user owns several properties, each property contains several rooms, each room holds several projects and each project several versions; every version has its generated images and product recommendations, with each recommendation pointing to a catalogue product, while each project may also hold chat sessions and their messages.

This hierarchy was an important product decision: a real renovation is not a single isolated image. A user may own more than one property, with several rooms per property, several proposals per room, successive revisions and versions for different budgets. The hierarchical structure helps the platform be perceived as a planning tool rather than merely an image generator. Access to properties and their children is mediated by a security-definer function that avoids policy recursion, and the data was backfilled idempotently so that legacy projects without a property were automatically linked to a default one.

3.3.3 Data dictionary and screens

To complement the data model, this subsection documents the main intake and product fields and the principal screens of the application. The intake (anamnesis) collects the data that guides the generation; the main fields and their purpose are listed below.

Table 10 – Intake fields

Field	Purpose	Status
Room type	Contextualise the layout and furniture	Concluded/partial

Dimensions	Improve the sense of scale	Partial
Ceiling height	Adjust spatial perception	Planned/partial
Doors and windows	Avoid unfeasible suggestions	Partial
Main use	Direct the room's function	Concluded/partial
Desired style	Guide the aesthetics	Concluded
Colour palette	Control the appearance	Partial
Mandatory items	Preserve the user's needs	Partial
Restrictions	Avoid incompatible solutions	Partial
Generic budget	Modulate the recommendations	Partial
Pets / allergies	Adjust materials and choices	Planned/partial
Room photo	Visual basis	Partial/concluded

Source: prepared by the author (2026).

The product catalogue uses standardised fields so that recommendations can be filtered, displayed and explained.

Table 11 – Product fields

Field	Description
Product name	Commercial name
Main category	For example sofa, bed or table
Subcategory	For example reclining, double or office
Store	Origin of the product
Current price	Value at the time of collection
Promotional price	When applicable
Dimensions	Item measurements
Predominant material	Wood, MDF, metal, fabric and so on
Colour / finish	Aesthetic information
Short description	Item summary
Link	Product URL
Associated style	To confirm
Compatible rooms	To confirm

Source: prepared by the author (2026).

The application is organised into a set of screens and routes, summarised below.

Table 12 – Main screens and routes

Area	Purpose
Landing page	Communicates the proposition to visitors
Sign-up / login	Authentication and access to the logged-in area
Dashboard	Entry point for the logged-in user
New project	The project-creation flow
Project result	Displays the generation and the project data
My renovations (history)	Access to previous projects
Properties	Organisation of properties
Rooms	Organisation of rooms
AI Designer	Conversational design experience
Retailer portal	Partner area
Condominiums	Condominium vertical
Layout Studio 3D	Spatial visualisation

Source: prepared by the author (2026).

3.3.4 APIs and functions

The server logic runs in serverless functions grouped by domain. The main groups are summarised below.

Table 13 – API function groups

Group	Main responsibilities	Status
Generation	Image generation, briefing interpretation, computer vision, furniture and mood extraction, anchor-object identification, embeddings and event tracking	Concluded/partial
Agents	Persona chat, comparative panel, debate, synthesis and conversation persistence	Prototype/partial
Image editing	Image edits and anchor/depth-based variations via the fallback provider	Partial
Collaboration	Routes for preference voting and mediation (not exposed in the main navigation)	Prototype
Context	Auxiliary data lookups, such as climate by location	Partial

Source: prepared by the author (2026).

3.3.5 Security and compliance

Because image generation is a costly operation that handles personal data and photos of homes, security and compliance were treated as first-order concerns.

Each user must access only their own data. This is enforced with token-based sessions, Row Level Security (RLS) on every database table, protected routes and a security-definer helper function that checks membership without causing policy recursion. An earlier defect, in which an administrative policy queried its own table and caused infinite recursion (returning an HTTP 500 error on every profile read), was fixed by moving the check into a security-definer function.

Generation endpoints are protected by per-IP rate limiting via a managed Redis service, with an in-memory fallback for local development, and by per-user usage quotas. Input validation guards against malicious prompts, oversized payloads, incomplete data and abusive uploads. For a public document, the specific limits and commercial usage rules are not disclosed.

RenovAÍ handles personal data, images of residential spaces and consumption preferences, which requires attention to the Brazilian General Data Protection Law (LGPD): data minimisation, an explicit purpose, access control, the possibility of deletion and transparency about the use of images.

Table 14 – LGPD measures

Measure	Status

Terms of use published	Concluded
Privacy policy published	Concluded
Data export (portability)	Concluded
Account/data deletion	Concluded
Consent log	Partial/concluded
Encryption of sensitive identifiers	Partial/concluded
Row Level Security on all tables	Concluded
Defined image-retention policy	Planned/to confirm
Notice of non-replacement of a professional	Partial/concluded

Source: prepared by the author (2026).

Because the system uses generative models, there is a risk of prompt injection, jailbreaks and adversarial inputs. Application-level controls are already in production (multi-bucket rate limiting, payload limits, token authentication, RLS, input validation and encrypted identifiers). Recommended hardening that is not yet validated includes separating the system prompt from user input, validating structured output, preventing the model from returning internal instructions, content filtering and a formal penetration test. No formal penetration test has been conducted yet.

The platform depends on a set of external services, summarised below with their criticality and mitigation.

Table 15 – External dependencies

Service	Role	Criticality	Mitigation
Generative-AI provider	Image generation and text/vision	High	Multi-model fallback to alternative providers
Managed database/auth/storage	Data, authentication and files	High	Standard, portable PostgreSQL; regular review
Serverless hosting	Application execution and deployment	High	Portable functions; alternative providers exist
Managed Redis	Rate limiting	Medium	In-memory fallback in development
Image-editing provider	Variations and edits	Medium	Optional; the main flow does not depend on it

Source: prepared by the author (2026).

4 Results and Validation

This chapter reports the results of the work: the solution delivered and its modules (4.1), the technical evaluation and testing (4.2), the market analysis (4.3), the validation carried out with real users in production (4.4) and an overview of the business model (4.5). The detailed business plan is provided in Appendix A. The chapter applies the frameworks introduced in Chapter 2 to the solution whose construction was described in Chapter 3.

4.1 Delivered Solution and Modules

This section documents the development of each module: what it does, the steps taken to build it, the main challenges and the current status. The product's maturity is not homogeneous: some modules are in production, others exist as prototypes, internal routes or documented concepts.

The authentication module lets users create an account, log in, access the logged-in area and save their projects; without it the experience would be disposable, with no history or continuity. The development steps were: defining the need for a logged-in area; choosing a managed authentication service; building sign-up and login screens; integrating the user session; protecting private routes; associating projects with the authenticated user; and manual testing of sign-up, login, logout and persistence. The main challenges were balancing early friction (asking for login too soon reduces conversion) with persistence and security (a user must never access another user's data). Status: concluded/partial.

The property, room and version hierarchy module organises projects the way a real home is organised, structuring the work as Property to Room to Project to Versions, so a user can keep, for example, a minimalist, an industrial and a budget version of the same bedroom. The steps were recognising that an isolated 'project' was limiting; defining the conceptual hierarchy; modelling the entities in the database; building screens to list properties, detail a property and show rooms; attaching generations to rooms; introducing the concept of versions; and integrating it with the history. The main challenges were UX complexity (a lay user must not get lost), data migration (legacy projects without a property/room) and making versions easy to recognise and compare. Status: partial/concluded. The concept is consolidated and is one of the foundations of the current version; advanced comparison and versioning are still partial.

The project-creation flow, or intelligent intake, is the heart of the product: the flow that lets a user start a renovation. The simplified flow is new project, room data, preferences, photo or description, generation, result and history. The steps were building an initial form, defining the minimum briefing fields, adding room type, desired style and a generic budget band, enabling upload or description of the room, sending the data to the AI layer, showing a processing state, rendering the result, saving it and linking it to the history. The intake collects, among other fields, room type, dimensions, main use, desired style, colour palette, mandatory items, restrictions and a photo. The main challenges were balancing a short form with a complete briefing, avoiding the need for the user to write a technical prompt, handling poor photos, keeping consistency between the briefing and the generated image, and preventing the generation from drifting away from the original room. Status: concluded/partial.

The image-generation module turns the briefing into a photorealistic or conceptual visualisation of the renovated room. It is one of the most important and most difficult components, because a generative model can alter geometry, proportions, doors, windows and elements that should stay fixed. The steps were testing image models, creating initial prompts, separating user data from internal instructions, defining visual styles, adding constraints such as room type and budget, testing perspective and floor-plan outputs, attempting to preserve the room's structure, comparing models, adjusting visual quality and integrating it into the site with storage and display of the result. Output types explored include a photorealistic perspective render (the main deliverable), a generated floor plan (a prototype, with coherence difficulties), before/after (planned), style variations and a visual moodboard. Figure 3 shows a representative result.

Figure 3 – RenovAÍ result: original room (left) and AI-generated renovation (right)

Source: RenovAÍ platform (renovai.ia.br), 2026.

The main technical challenges were geometry preservation (the model may move a window, change proportions or invent doors), coherence between the floor plan and the perspective (two individually good images can be mutually inconsistent), scale control (suggested furniture may not fit the real room) and aesthetic quality (making the image look premium and realistic rather than generically 'AI-made'). Because prompt engineering is a differentiator, the specific techniques are kept generic here; at a high level the project used a structured briefing, a separation between mandatory constraints and aesthetic preferences, staged generation, fallback between models, reference images and post-hoc validation of the output. Status: concluded/partial. Generation works as the core of the product, but geometric fidelity, scale and automatic plausibility validation are not fully solved.

The furniture-recommendation module turns the image or briefing into a list of suggested furniture, ideally compatible with the room, style and budget, so that RenovAÍ delivers not only an inspirational image but also an understanding of which products could compose the result. The steps were defining the relevant furniture categories, collecting and structuring an initial catalogue, creating a CSV of products, inserting products into the database, defining standardised fields, testing LLM-based recommendation, experimenting with embeddings, associating products with room types, generating a textual recommendation list and partially integrating it with the project result. Product fields include name, main and sub category, store, current and promotional price, dimensions, predominant material, colour/finish, a short description and a link. The main challenges were price and stock updates, standardisation across stores, missing dimensions, the gap between textual similarity and real compatibility, subjective style classification, linking a real product to the generated image, and scale. Status: partial/prototype. Essential to the differentiator but still needing end-to-end validation, automatic catalogue updates, link checking and a stronger connection between the generated image and real products.

The history lets the user access previous renovations, review generated images and compare ideas. The steps were defining the need for persistence, associating projects with the user, building a 'my renovations' page, listing projects, showing cards with the main data, opening a project's detail and integrating it with the property/room/version structure. The main challenges were avoiding a confusing history, displaying images without slowness, organising versions and preventing improper access to another user's projects. Status: concluded/partial. A fundamental part of the MVP; future improvements include filters, search and a more visual organisation.

PDF export was designed to turn a project's result into a shareable document containing a summary of the room, the generated image, the chosen style, the suggested furniture list, notes, a generic estimate, links and next steps. The considered steps were defining the template, selecting the project data, inserting the generated image and the product list, organising generic totals, adding a limitations notice, generating the file, allowing download and testing on mobile and desktop. The main challenges were keeping links up to date, avoiding the appearance of an executive technical document, dealing with changing prices and generating a PDF with heavy images. Status: partial/to confirm. Planned and discussed; the final implementation should be confirmed in the repository and described as partial until formally tested.

The personas module simulates different professional perspectives on a room (for example, functionality, aesthetics, budget, ergonomics, lighting, circulation and style), so the user sees trade-offs rather than a single answer. The steps were defining the idea of multiple personas, creating agent profiles, defining roles and evaluation criteria, implementing a panel of agents, testing a debate between personas, generating a final synthesis and exploring integration with the AI Designer. The main challenges were excessive complexity and cognitive cost for a lay user, repetition between agents, artificial conflict, and turning the debate into a concrete decision. Status: prototype/partial. The concept and a debate/panel were explored; the main pending item is turning the debate into simple, actionable recommendations.

The AI Designer is a conversational experience in which the user interacts with agents to build or review a project, allowing a more open conversation than the form-based flow. It can act as a briefing assistant, a style consultant, an alternative generator and a mediator between conflicting preferences. The steps were defining the conversational-designer concept, creating a main agent and a specialist agent, implementing the chat, integrating it with floor-plan or textual input, testing free conversations, creating a panel of agents and attempting to convert the conversation into a structured briefing. The main challenges were maintaining long conversational context, turning conversation into structured data, avoiding generic answers, integrating the chat with image generation and reducing hallucinations. Status: prototype/partial. One of the most promising fronts; the multi-agent version is experimental, while a version with few agents is more viable in the short term.

The Retailer Portal was conceived as an area for partners to register and manage products, connecting RenovAÍ's catalogue to real furniture. Possible features include store registration, product registration, price and stock updates, image upload, categorisation and integration with recommendations. The considered steps were defining the retailer vertical, modelling the retailer and per-retailer product entities, designing a panel, discussing catalogue import and planning integration with the recommendation flow. The main challenges were the constant change of a store's catalogue, standardisation across retailers, data quality, integration with the generated room and operational scale. Status: draft/planned. A product front rather than a finished module; an initial interface or documentation may exist, but full operation is not consolidated.

Layout Studio 3D aims to represent rooms three-dimensionally, letting the user arrange furniture, view proportions and bring the project closer to a spatial experience. Experiments were carried out with A-Frame and 3D assets, including local tests of multiple rooms and GLB/OBJ models; the production application also uses Three.js with React Three Fiber. The steps were researching web 3D libraries, building a simple 3D scene, inserting walls, floor and furniture, testing ready-made assets, adjusting scale and height, testing navigation and evaluating integration with the AI-generated layout. The main challenges were scale, asset quality and size, performance on phones, the UX of editing in 3D for a lay user, and the difficulty of converting a briefing into a 3D scene. Status: prototype. It validated possibilities but is not fully integrated into the main flow.

The moodboard is a visual composition of style references, colours, materials, textures, furniture and atmosphere, helping the user understand the aesthetic direction before or after the render. The considered steps were defining its role in the flow, extracting the style and palette from the briefing, generating or selecting visual references, organising blocks of colour, materials and key furniture, connecting it to the final render and allowing comparison between styles. The main challenges were avoiding the misuse of protected images, connecting references to real products, avoiding a generic moodboard and preserving a Brazilian identity. Status: partial/planned. Coherent with the proposition; the final implementation should be confirmed and may be treated as a future or experimental feature.

Analytics makes it possible to understand how users use the product, where they drop off and which modules generate the most engagement. For a public document, sensitive acquisition or monetisation figures are avoided; the focus is on the existence and technical planning of events. Useful events include project started, briefing completed, generation started/completed/failed, project saved, version created, product clicked, PDF exported, chat started, room created and property created. The main challenges were privacy (events must not expose personal content), instrumentation (every screen must record events), interpretation, LGPD transparency and event quality. Status: partial/to confirm. Basic events or logs may exist, but a complete metrics-oriented architecture should not be assumed.

The institutional landing page communicates the proposition to visitors, explaining the problem, the solution and how it works, and directing the user to start a project; it has gone through continuous visual evolution and may still require copy, branding, responsiveness and performance refinement (status: concluded/partial). Two further fronts were explored but are not finished products: a condominium vertical

(Condominium Retrofit), applying the platform to common areas such as halls, gyms and party rooms at a conceptual level and without structural responsibility (status: concept/partial vertical); and preference-mediation ideas for people who decide together, available as a hidden internal route rather than a finished feature (status: concept/prototype). A full marketplace (checkout, cart, payment, order and reputation management) was not implemented and remains a future direction.

4.2 Technical Evaluation and Testing

Testing was predominantly manual and exploratory, appropriate to a lean, fast-moving project, and is documented here honestly, including the testing that was planned but not implemented.

Manual flow tests were run throughout development to validate that screens and flows worked (sign-up/login, project creation, form completion, sending data to the AI, image return, result display, saving, navigation to the history and opening a previous project). Image generation was tested with real or semi-real cases across different rooms, styles and requests, which surfaced problems such as spatial inconsistency, undue changes to the structure, excessive decoration, incompatible furniture and visually good but impractical results. Catalogue experiments (CSV, database seed and recommendation) revealed the need to standardise categories, inconsistent store data and the need to validate embeddings and semantic search. The 3D/AR experiments with A-Frame revealed scale, performance, asset-quality and integration difficulties. Exploratory validation with users and potential users was also carried out; any percentages or samples used in this work are taken directly from the original research and cited accordingly, and no metrics are invented here. The platform has generated 67 projects in production with completed generation, validating the end-to-end pipeline (briefing interpretation, image generation, furniture extraction and storage).

There is no robust automated suite of unit, integration and end-to-end tests covering the whole application. For honest documentation, the status of each test type is summarised below.

Table 16 – Status of automated testing

Test type	Status
Unit tests	Not implemented / to confirm
Integration tests	Not implemented / to confirm
End-to-end tests	Not implemented / to confirm
Automated API tests	Not implemented / to confirm
Automated image-generation tests	Not implemented

Visual-regression tests	Not implemented
Automated accessibility tests	Not implemented
Automated security tests	Not implemented
CI/CD with a full suite	Not implemented / to confirm

Source: prepared by the author (2026).

Several types of testing were considered but not implemented, and are listed here as part of an honest account of the project's maturity and as a basis for future work.

Table 17 – Planned but not implemented testing

Test	What it would cover	Status
Unit tests	Isolated functions: form validation, briefing normalisation, AI payload building, product formatting, permissions	Planned/not implemented
Integration tests	Frontend-backend, backend-database, backend-AI, auth and private routes, generation and storage	Planned/not implemented
End-to-end tests	Full flows: sign-up to project to generation to history; login to property to room to version; project to generation to PDF	Planned/not implemented
Load and stress tests	Many simultaneous generations, response time, external-API failures, behaviour at usage limits	Planned/not implemented
Security penetration test	Improper project access, auth flaws, malicious upload, data exposure, prompt injection, key leakage	Planned/not implemented
Accessibility tests	Contrast, keyboard navigation, form labels, alt text, screen readers, error states	Planned/not implemented
Cross-device/browser tests	Chrome desktop, Safari iOS, Chrome Android, Edge, Firefox, different screen sizes, mobile upload	Partial/not formalised
Closed beta	Comprehension of the proposition, image quality, usefulness of recommendations, sign-up friction, perceived trust	Planned/not implemented
A/B tests	Short vs complete form, photo before vs after briefing, landing CTA, chat vs form, one vs multiple versions	Planned/not implemented

Source: prepared by the author (2026).

Quality assurance was mostly manual and iterative, following a recurring cycle: implementation, a manual test, identification of bugs, a fix (with AI assistance), a new manual test, visual validation, deployment and a further test in production.

Table 18 – Observable bug types

Type	Example
Visual	Broken layout on mobile
State	A project does not appear after generation
AI	The image ignores the briefing
Database	An incorrect project/user relation
Upload	A photo does not load
Performance	A screen is slow with no feedback
Deployment	A build fails due to a dependency/configuration
Route	A page is accessible directly without context
UX	The user does not understand the next step

Source: prepared by the author (2026).

4.3 Market Analysis

This section reports the market analysis that supports the business case: the strategic assumptions and hypotheses (4.3.1), the market sizing (4.3.2) and the competitive landscape (4.3.3).

4.3.1 Market assumptions and hypotheses

The product was designed around three explicit, falsifiable hypotheses, each to be confirmed or refuted with proprietary data collected from real usage.

The problem hypothesis is that most people planning a renovation face a real pain point, the fear of buying wrong, and do not hire a professional, given the cost; the assumption is that this public will adopt a free tool that delivers visual certainty before they spend.

The solution hypothesis is that a pure generative-AI solution (instant, mobile-first, with a furniture list linked to real stores) is a better way to solve the problem than the existing alternatives, namely human-in-the-loop services that cost from R$ 149 and take days, or web-only AI tools without a shoppable list.

The value hypothesis is that the revenue model is acceptable to users: the visualisation is free, monetised indirectly through commissions on furniture the user would buy anyway, with optional paid plans for heavier users.

4.3.2 Market sizing

Market size was estimated bottom-up, with each factor traceable to a published primary source.

The market size is decomposed into three layers: TAM, SAM and SOM. The Total Addressable Market is annual spending by Brazilian urban households on home goods. The Serviceable Available Market restricts it to households with purchasing power and service consideration. The Serviceable Obtainable Market is RenovAÍ's conservative three-year potential, an internal assumption to be validated.

Table 19 – Market sizing (TAM, SAM, SOM)

Layer	Value	Methodology / source
TAM	R$ 92 billion/year	67.7 million urban households (FJP, 2024) × R$ 1,362 average home-goods spend (ABCasa/IEMI, 2024)
SAM	R$ 30.8 billion/year	TAM × 45.7% (classes A to C1, ABEP, 2024) × 73% (would consider an architect, CAU/Datafolha, 2022)
SOM (year 3)	R$ 8.5 million/year	0.5% penetration of the SAM × hybrid ARPU of R$ 75/year, an internal assumption to validate

Source: prepared by the author (2026), based on the cited sources.

The Brazilian context reinforces the opportunity. The home-goods sector moved approximately R$ 102 billion in 2024 (ABCasa/IEMI, 2024), and a large share of households live in homes with some form of inadequacy: the Fundação João Pinheiro estimates millions of urban households with qualitative housing inadequacy and a quantitative housing deficit (FJP, 2024). Combined with the fact that 82% of homes are built without a qualified professional (CAU/Datafolha, 2022), this points to a large base of people who renovate the home they already have, largely on their own, precisely the public RenovAÍ serves.

Table 20 – Public context indicators

Indicator	Value	Source
Home-goods sector (2024)	~R$ 102 billion	ABCasa/IEMI, 2024
Homes built without a qualified professional	82%	CAU/Datafolha, 2022
Average residential architecture cost	R$ 55 to R$ 140 per square metre	GetNinjas, 2024; Habitissimo, 2024
Urban households with housing inadequacy/deficit	Millions of households	FJP, 2024

Source: prepared by the author (2026), based on the cited sources.

In terms of customer segmentation, RenovAÍ serves several segments, from mass-market renovators to higher-income owners of compact properties, in addition to local artisans as future marketplace suppliers, as summarised in the table below.

Table 21 – Customer personas

Persona	Profile	Job to be done
A: Decided Renovator	Adult, middle class, renovating a room	Certainty before spending
B: Urban Dweller	Young adult, first apartment	Translate inspiration into a decision
C: Compact High-End Owner	Young high-income, compact flat	Quality result fast, no waiting
D: Local Artisan	Artisan, future marketplace supplier	Reach buyers with a defined project

Source: prepared by the author (2026).

Persona A, the decided renovator, is the core audience: an adult, often middle class, who is going to renovate or furnish a room and whose main pain is the fear of buying wrong, having never hired an architect; they need certainty before spending and decide carefully, with high price sensitivity.

Persona B, the urban dweller, is a younger adult, frequently in a first apartment, who has a clear aesthetic vision but struggles to articulate it for execution; they are more experimental and often share the result on social media, seeking to translate diffuse inspiration into a concrete decision.

Persona C, the owner of a compact high-end property, is a young, higher-income user living in a small flat in a valued neighbourhood; sensitive to time rather than price, they want a quality result quickly, without waiting weeks for a project.

Persona D, the local artisan, is not a design buyer but the supplier of the future marketplace: a carpenter, artisan or product designer who lacks a qualified sales channel and would benefit from reaching buyers who already have a defined visual project.

4.3.3 Competitive analysis

A mapping conducted in May 2026 identified 27 relevant players. The Brazilian AI-assisted interior-design market is more crowded than a first read suggests, but the segments differ.

Direct Brazilian competitors include both AI-only tools and human-assisted services (such as Arquiteto de Bolso, DecoraGPT, IAdecora and Archie). International references that frame the category include Wayfair (with its Decorify/Muse AI tools, but absent from Brazil), Houzz, RoomGPT and Havenly, while the closure of the pure-human Modsy illustrates that human-in-the-loop models struggle to scale.

Table 22 – Competitive comparison (overview)

Criterion	RenovAÍ	AI-only tools	Human-assisted services
Price	Free (MVP)	Free / paid	From R$ 149
Delivery	AI, under a minute	AI, minutes	Human, days
Localisation (PT-BR, BRL)	Yes	Varies	Yes
Mobile-first experience	Yes	Mostly web	Mostly web

Source: prepared by the author (2026).

Five international references frame the possible space for the category and help calibrate expectations, as summarised in the table below.

Table 23 – International benchmarks

Reference	Country	Model	Lesson for RenovAÍ
Wayfair (Decorify/Muse)	USA	Retailer with integrated generative AI; not present in Brazil	Validates the 'AI visualisation + shoppable products' category; the Brazilian window is open
Houzz	USA	Marketplace, tools and a professional community	Shows the possible ceiling when community and catalogue become the moat
RoomGPT	USA (indie)	Pure-AI, web-only, freemium	Proves demand for pure-AI generation, with a positive exit
Havenly	USA	AI plus human plus a marketplace	The highest-monetisation model; combines speed with curation

Modsy	USA (closed 2022)	Pure-human 3D rendering	Shows that human-in-the-loop models struggle to scale

Source: prepared by the author (2026).

RenovAÍ positions itself consciously between a pure-AI tool (speed) and a curated marketplace, without the marginal cost of human designers and adapted to the Brazilian market, a combination not occupied by the international references, which do not serve Brazil natively.

In terms of competitive advantage, generative AI itself is an increasingly commoditised resource; RenovAÍ differentiates through deep localisation for the Brazilian market (prices in reais, local brands, native Portuguese), a mobile-first experience and ease of use for non-experts, supported by a low operating cost.

4.4 Validation and Results

Validation followed a lean, evidence-based approach, using the product in production with a controlled user base to test the business hypotheses rather than to claim market traction.

In terms of methodology, the MVP was released in production and instrumented with funnel analytics (sign-up, wizard completion, successful generation, sharing, PDF export) under explicit LGPD consent, with UTM attribution by channel.

As of May 2026 the platform had 20 registered users and 8 users who generated at least one project, totalling 67 completed generations. The activation rate (sign-up to first generation) was 40%, identifying the activation funnel as the main bottleneck. These figures constitute deliberate technical validation, not statistical market traction.

Validation refined the revenue assumptions and led to the introduction of optional paid plans alongside the commission model, while preserving the core thesis rather than abandoning it.

Table 24 – Key performance indicators monitored

KPI	What it measures	How to measure
Registered users	Reach	Database
Active users	Real product usage	Database
Activation rate	Sign-up to first generation	Funnel analytics
Click-through on products	Purchase intent	Link tracking
Conversion to paid plan	Monetisation	Subscription data

Source: prepared by the author (2026).

Several categories of risk were considered. The financial risk is that the revenue model still requires validation with proprietary data, mitigated by a lean cost base and optional paid plans. The technological risk is the dependence on third-party AI services, mitigated by provider redundancy in the pipeline. The competitive risk is that a well-funded incumbent could enter, mitigated by deep localisation and product focus. Finally, the legal risk concerns data protection (LGPD) and the tax treatment of revenue, mitigated by the basic compliance already implemented and a planned legal review. The main technical risks are detailed in the table below.

Table 25 – Technical risk analysis

Risk	Probability	Impact	Mitigation
AI provider price or policy change	Medium	High	Multi-model architecture with fallback
External API unavailability	Medium	High	Provider redundancy and error handling
Geometric inconsistency in generation	High	Medium	Structured briefing, staged generation, future validation
Outdated catalogue/links	High	Medium	Planned automatic updates and link checking
Lack of automated tests	High	Medium	Planned test suite; manual QA in the interim
Security / prompt injection	Medium	High	Application controls in production; planned penetration test
Exchange-rate impact on AI cost	High	Medium	Structural fall in AI cost; provider flexibility

Source: prepared by the author (2026).

4.4.1 Technical challenges and lessons learned

The development surfaced a set of technical challenges whose resolution shaped the product. They are summarised below, each with the main lesson learned.

Visual quality alone is not enough: an image can look sophisticated and still be poor for planning if it changes the structure of the room, suggests furniture that does not fit, ignores constraints, invents elements or exceeds the budget. The lesson is that the AI must be evaluated by usefulness, not only by aesthetics.

Preserving the geometry of the room is difficult: most generative models tend to embellish a space but may alter its structure. Mitigations considered include a more structured briefing, collecting measurements, using a reference image, staged control and post-hoc validation. Resolution status: partial.

Generating a coherent floor plan and perspective is harder than generating a single image, because the system must understand the space consistently. The lesson is that a floor plan requires a more structured approach, possibly with an intermediate geometric model, rather than purely visual generation.

Recommending real products requires far more than listing attractive items: it must consider measurements, style, price, availability, store, material, colour, lead time, compatibility with the room and substitutes. The lesson is that the catalogue must be treated as a data system, not merely as text.

RenovAÍ cannot depend on users writing detailed commands; the product must extract information through the interface, questions and guided choices. The lesson is that the intake is as important as the AI model itself.

Agents and personas enrich the analysis but can also make the experience complex; a lay user may not want to follow a debate between ten specialists. The lesson is that a multi-agent approach should be used internally or in summary form, delivering an actionable synthesis.

RenovAÍ must avoid any promise of a structural technical project, which directly influenced the product's positioning. The lesson is that explicit limits increase safety and credibility.

Many fronts emerged during the project (3D, AR, retailer, condominiums, personas, chat, catalogue, marketplace); the challenge is to keep focus. The lesson is that the product's core must remain clear: helping the user visualise, decide and plan a light renovation.

AI services change price, availability, latency, quality and policy, which makes the architecture vulnerable if it relies on a single provider. The lesson is that the application must abstract AI providers and allow fallback, without exposing sensitive detail.

Because the project evolved quickly, not every decision was documented when it was made. The lesson, for this work, is the need to reconstruct the trajectory honestly, marking status and uncertainties.

4.4.2 Features and ideas not implemented

An honest account of the project includes the features that were ideated, prototyped or abandoned. They are documented here both for transparency and as a basis for future work; the table separates what was done from what was not.

Table 26 – Features and ideas not (fully) implemented

Idea	What was done	What was not done	Status
AR/VR	Technical experiments with A-Frame, local 3D scenes and assets	Full integration into the main flow, stable mobile AR, a real 3D catalogue	Technical prototype
Condominium Retrofit	Vertical ideation, a possible dedicated section, use-case discussion	A complete system for managers, a collective-approval flow, technical budgeting	Concept/partial
Iterative AI Designer	Conversational chat and agents explored	A robust image-editing loop, consistent geometry across iterations, a visual diff between versions	Partial/prototype
Persona voices	Textual personas and debate explored	Real audio, synthetic voices, a multimodal experience	Idea/not implemented
Full marketplace	Initial catalogue, product recommendation, retailer-portal idea	Checkout, cart, payment, order and reputation management, catalogue reconciliation at scale	Not implemented
End-to-end semantic search	Embedding experiments, an initial product base, recommendation tests	Systematic relevance validation, benchmarking against filters, automatic base updates	Partial/not validated
Robust anti-prompt-injection	Consideration of validations, conceptual separation of internal instructions and user input	A formal penetration test, an adversarial-attack suite, complete server-side validation, an external audit	Partial/not validated
Native mobile app	Mobile-first web product	An iOS/Android native application	Planned/not implemented

Source: prepared by the author (2026).

4.5 Business Model

At a high level, RenovAÍ follows a common digital business model for a free-to-use consumer application: the core experience is offered at no cost to maximise adoption, and revenue is expected to come from indirect, low-friction sources rather than from charging the end user for the visualisation itself. The mechanisms considered are the standard ones described in Section 2.6: commissions or affiliate revenue on furniture the user would buy anyway, optional paid plans for heavier or professional users, and a possible future marketplace connecting suppliers to buyers who already have a defined visual project.

The cost structure is lean and cloud-native, with a low fixed cost and a variable cost per generation that has been falling alongside the broader decline in generative-AI prices; the model carries no inventory or logistics. Financial feasibility therefore depends less on margin per unit and more on validating user adoption and conversion with proprietary data, for which the low cost base provides runway. A consolidated, deliberately generic business plan (market analysis, business model canvas, go-to-market and feasibility) is presented in Appendix A; the figures there are illustrative and conservative rather than a disclosure of the operation's internal strategy.

5 Conclusion

This work developed and validated RenovAÍ, a generative-AI platform that lets any Brazilian visualise a renovated room and obtain a furniture list for free, and elaborated the corresponding business plan. The objectives were met: a functional MVP is in production, validated technically with 67 generated projects; a revenue model was defined; and a complete business plan, including market sizing, competitive analysis and financial feasibility, was produced.

More precisely, the balance between what was proposed in Section 1.3 and what was delivered can be examined objective by objective. The technical objectives were fully met; the business-related objectives were met at the level of design and initial evidence, with full market validation deliberately left as future work given the controlled scale of the test base. The table below makes this comparison explicit, including the points that remain partial and the reason for each.

Table 27 – Comparison between proposed and delivered objectives

Objective (proposed)	What was delivered	Status and justification
Build a functional MVP generating a photorealistic visualisation and a furniture list	MVP in production at renovai.ia.br, generating images and shoppable furniture lists from a briefing	Met: the core flow works end to end in production
Validate the product technically with real users	67 projects generated by real users; functional and manual QA testing	Met technically, but with a small, controlled base; broad market validation remains future work
Define a revenue model	Commission/affiliate model with optional paid plans and a future marketplace	Met as a design: user conversion not yet validated with proprietary data
Elaborate a complete business plan	Market sizing, competitive analysis, business model and feasibility (generic version in Appendix A)	Met: kept deliberately conservative and generic in the public version

Source: prepared by the author (2026).

The central finding is that the solution demonstrates technical feasibility and initial robustness, and that the business model appears feasible, contingent on validating user conversion with proprietary data. The platform's lean, cloud-native cost base provides runway for that validation.

Future work includes formalising the company, validating the revenue assumptions with real data, expanding the product across platforms, and growing the user base. The technical roadmap below complements these steps.

5.1 Future technical work

Beyond the business steps, a clear technical roadmap emerged from the development. The priorities below are described without business strategy.

Before expanding new features, it is advisable to consolidate the real folder structure, the database entities, the relationship between projects, rooms and versions, the AI layer, logging, permissions, error handling and deployment. Priority: high.

The platform should evolve to include automated tests, prioritised as follows: unit tests of critical functions (high), integration tests with the database (high), end-to-end tests of the main flow (high), security/penetration testing (high), visual-regression and accessibility tests (medium) and load/stress tests (medium).

Possible technical paths include mandatory collection of measurements, the use of a simplified floor plan, detection of structural elements, segmentation of the room, layout control via JSON, automatic object validation, comparison between the original and generated image, and staged generation.

Instead of generating an image directly from text, the system could first produce a structured layout that acts as a bridge between intent, image, products and 3D (recording, for example, the room type, its dimensions, the fixed elements such as doors and windows, and a list of furniture with the type, position and size of each piece) and only then render the final image from that structured description.

Next steps include normalising the catalogue, validating links, updating availability, classifying products by style, considering dimensions, improving room matching, combining traditional filters with semantic search and explaining each recommendation.

Further work includes implementing version comparison (side by side, with the main differences and a generic cost level); strengthening the AI Designer (a conversational briefing, project memory, adaptive questions and persona synthesis); improving observability (generation and error logs, response time, a technical-failure rate and exception tracking); maturing security and LGPD (reviewing the privacy policy and terms, image-retention limits, reinforced RLS, upload validation and prompt-injection testing); improving accessibility (input labels, contrast, keyboard navigation, focus states, alt text and screen-reader support); improving performance (image compression, lazy and skeleton loading, result caching and timeout handling); integrating 3D progressively (from a structured 2D layout to a simple visualisation and an optional AR mode); and producing technical documentation (architecture, data model, generation flow, an internal prompt guide without exposing differentiators, a security guide, a test guide, a module map, discarded decisions and a deployment manual).

5.2 Limitations of the work

This work has limitations that should be made explicit. The validation was deliberately technical, with a controlled and small user base, so the conversion and market hypotheses still require broader testing; the automated-test coverage is minimal, which increases the risk of regressions; several modules remain at the prototype or concept stage, with non-homogeneous maturity; the architecture evolved over time, which makes some older decisions harder to trace; and the platform depends on third-party AI services whose price, availability and behaviour may change. These limitations do not invalidate the contribution of the work (a functional MVP, a documented architecture and process, and an honest map of what was built, planned and discarded), but they delimit the scope of the conclusions and orient the future work described above.

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Appendices

This section consolidates supporting material for the work: a generic business plan, a summary of the modules and their status, a glossary of terms and a list of items to be confirmed against the repository.

Appendix A – Business Plan (generic overview)

The business plan below is intentionally generic and conservative. It illustrates a common, viable model for a free-to-use consumer application and does not disclose the internal commercial strategy, specific negotiated rates or detailed unit economics of the operation.

RenovAÍ addresses the broad Brazilian market of people who renovate, decorate or reorganise the home they already have. The opportunity, the market sizing (TAM, SAM, SOM) and the competitive landscape are presented in Section 4.3; in summary, the category is large, growing and under-served by tools that combine instant generative visualisation with deep localisation and a shoppable furniture list.

Table 28 – Business Model Canvas (generic overview)

Block	Generic content
Customer Segments	General-public renovators; potentially retailers and developers (B2B)
Value Proposition	See your renovation before buying: photorealistic visualisation and a furniture list, free and in minutes
Channels	Mobile-first web; organic search and social media; referral; partnerships
Customer Relationships	Self-service product with a conversational assistant and support by e-mail
Revenue Streams	Commission/affiliate on referred sales; optional paid plans; possible future marketplace
Key Resources	Generative-AI pipeline; product and engineering; user base with saved projects
Key Activities	Product development; AI integration; partnership onboarding
Key Partnerships	Cloud and AI providers; furniture retailers; potential funding agencies
Cost Structure	Lean, cloud-native fixed cost; variable cost per generation; no inventory or logistics

Source: prepared by the author (2026), generic overview.

The go-to-market is fully digital, with no human sales force: acquisition relies on organic channels and partnerships, with paid media introduced only once revenue can sustain the cost of acquisition. Feasibility rests on a low, cloud-native fixed-cost base and a falling variable cost per generation; the figures used for planning are illustrative and conservative, and full financial validation depends on proprietary conversion data collected in production. Initial funding was bootstrapped by the founder, with non-dilutive public grants under consideration.

Appendix B – Consolidated module and status summary

The table below consolidates the modules and fronts of RenovAÍ with their current maturity.

Table 29 – Modules and status

Module	Status	Note
Landing page	Concluded/partial	Public site live and evolving
Authentication	Concluded/partial	Required for the logged-in area
New project	Concluded/partial	Main MVP flow
Project result	Concluded/partial	Displays the generation and data
History	Concluded/partial	Allows returning to projects
Property	Partial	Higher-level organisation
Room	Partial	Organisation by room
Versions	Partial	Fundamental for comparison
Image generation	Concluded/partial	Core of the product
Floor plan	Prototype/partial	Coherence still a challenge
Furniture recommendation	Partial/prototype	Needs validation
Catalogue	Partial/prototype	Base structured, maintenance pending
Budget PDF	Partial/to confirm	Verify final implementation
AI Designer	Prototype/partial	Conversational
Personas	Prototype	Debate/evaluation
Retailer Portal	Draft/planned	Not consolidated
Condominiums	Concept/partial	Explored vertical
Layout Studio 3D	Prototype	Technical experiments
AR/VR	Prototype	Not a final product
Moodboard	Partial/planned	Verify implementation
Analytics	Partial/to confirm	Robustness not confirmed
Marketplace	Not implemented	Concept/future direction
Native app	Not implemented	Possible future

Source: prepared by the author (2026).

Appendix C – Glossary of terms and acronyms

Table 30 – Glossary

Term	Meaning
AI	Artificial intelligence
LLM	Large language model
MVP	Minimum viable product
RLS	Row Level Security (per-row database access control)
LGPD	Brazilian General Data Protection Law
API	Application Programming Interface
CDN	Content Delivery Network
TAM/SAM/SOM	Total / Serviceable Available / Serviceable Obtainable Market
SWOT	Strengths, Weaknesses, Opportunities and Threats
BMC	Business Model Canvas
UX	User experience
CSV	Comma-separated values (a tabular data format)
Embedding	A numerical vector representing the meaning of a text or image
Prompt injection	An attack that manipulates a model through crafted input
Fallback	An automatic switch to an alternative provider on failure
GLB / OBJ	3D model file formats

Briefing	The structured set of the user's preferences and constraints

Source: prepared by the author (2026).

Appendix D – Items to be confirmed in the repository

Before turning this report into the final version, a set of items should be confirmed against the current repository: the exact current stack, the real folder structure and the technologies effectively in production; the real database tables, route names and Row Level Security policies; the status of PDF export, of the moodboard and of analytics; the automated-test coverage, if any; the existence of terms of use and a privacy policy; logging and observability; the active AI integrations; and which modules are public, private or accessible only by a direct URL.

Annexes

No annexes (third-party documents) are included in this version.